

LF Denoising Network		
Alias	Layer	Operations
Primary Feature Extraction	LF To ViewStack	concat
	PFE_1	conv2d (n32 k3 s1)
	PFE_2	conv2d (n32 k3 s1 dila1)-> conv2d (n32 k3 s1 dila2)-> conv2d (n32 k3 s1 dila4)->concat
	PFE_3	conv2d (n32 k3 s1)
Dual-attention Module	PFE_4	conv2d (n32 k3 s1)->relu-> conv2d (n32 k3 s1)
	VA1 (View Attention) Input	add (PFE_3,PFE_4)
	VA1_FC1	transpose
	VA1_FC2	GlobalMeanPooling ->conv2d (n16 k3 s1) -> conv2d (n16 k1 s1)
	VA1_value	GlobalMaxPooling ->conv2d (n16 k3 s1) -> conv2d (n16 k1 s1)
	VA1 Output	merge(FC1,FC2)->sigmoid
	VA2 Input	multiply (VA1_value,VA1)
	VA2_FC1	conv2d (n32 k3 s1)->relu-> conv2d (n32 k3 s1)
	VA2_FC2	conv2d (n32 k3 s1)->relu-> conv2d (n32 k3 s1)
	VA2_value	merge (VA1 Input, VA1 Output)
	VA2 Output	GlobalMeanPooling ->conv2d (n16 k3 s1) -> conv2d (n16 k1 s1)
	VA results	GlobalMaxPooling ->conv2d (n16 k3 s1) -> conv2d (n16 k1 s1)
	CA1 (Channel Attention) Input	merge(FC1,FC2)->sigmoid
	CA1_FC1	multiply (VA1_value,VA1)
	CA1_FC2	conv2d (n32 k3 s1)->relu-> conv2d (n32 k3 s1)
	CA1_value	conv2d (n32 k3 s1)->relu-> conv2d (n32 k3 s1)
	CA1 Output	merge (VA1 Input, VA1 Output,VA2 Output)
	CA2 Input	transpose
	CA2_FC1	GlobalMeanPooling ->conv2d (n16 k3 s1) -> conv2d (n16 k1 s1)
	CA2_FC2	GlobalMaxPooling ->conv2d (n16 k3 s1) -> conv2d (n16 k1 s1)
	CA2_value	merge(FC1,FC2)->sigmoid
Feature Fusion & Output	CA2 Output	multiply (CA1_value,CA1)
	CA results	conv2d (n32 k3 s1)->relu-> conv2d (n32 k3 s1)
	F_0	conv2d (n32 k3 s1)->relu-> conv2d (n32 k3 s1)
	F_1	conv2d (n8 k3 s1)->relu-> conv2d (n16 k3 s1)
	F_2	merge(F_0,F_1)
	F_3	conv2d (n8 k3 s1)->relu-> conv2d (n16 k3 s1)
	Output	merge(F_2,F_3)
	Conver to SAls	conv2d (n1 k3 s1)
		concat

LF De-aliasing Network		
Alias	Layer	Operations
Input	conver to Macron	concat
Angular Branch	PFE_A1	conv2d (n64 k15 s15)
	F1_S1	conv2d (n14400 k1 s1)->subpixel(r=15)
	F1_S2	concat (PFE_S1,F1_S1)->conv2d (n64 k3 s1 dila15)->Relu
	F1_S3	merge(PFE_A1,F1_A2)
	F1_S1_1	conv2d (n14400 k1 s1)->subpixel(r=15)
	F1_S2_1	concat (PFE_S1,F1_S1_1)->conv2d (n64 k3 s1 dila15)->Relu
	F1_S3_1	merge(PFE_A1,F1_A2_1)
	F2_A1	conv2d (n64 k15 s15)
	F2_A2	concat (F1_A3,F2_A1)->conv2d (n64 k3 s1)->Relu
Spatial Branch	F2_A3	merge(F1_A3,F2_A2)
	PFE_S1	conv2d (n64 k3 s1 dila15)
	F1_A1	conv2d (n64 k15 s15)
	F1_A2	concat (PFE_A1,F1_A1)->conv2d (n64 k3 s1)->Relu
	F1_A3	merge(PFE_A1,F1_A2)
	F1_A1_1	conv2d (n64 k15 s15)
	F1_A2_1	concat (PFE_A1,F1_A1_1)->conv2d (n64 k3 s1)->Relu
	F1_A3_1	merge(PFE_A1,F1_A2_1)
	F2_S1	conv2d (n14400 k1 s1)->subpixel(r=15)
Feature Fusion	F2_S2	concat (F1_S3,F2_S1)->conv2d (n64 k3 s1 dila15)->Relu
	F2_S3	merge(F1_S3,F2_S2)
	L1_A	concat(F1_A3_1,F2_A3)->conv2d(n64 k1 s1)->Relu->conv2d(n14400 k1 s1)->subpixel
Output	L1_S	concat(F1_S3_1,F2_S3,L1_A)->conv2d (n64 k3 s1 dila15)->Relu
	L2_S	merge(PFE_S1,L1_S)
Output	Output	conv2d (n25 k3 s1 dila15)->subpixel (r=5)->conv2d(n1 k1 s1)

3D Reconstruction Network		
Alias	Layer	Operations
Input	Inputlayer	
Spatial Feature Extraction	reshape	Space to Depth
	I0	conv2d(n128 k7 s1)
	I1	subpixel(r=2)->conv2d(n64 k3 s1)
	I2	subpixel(r=2)->conv2d(n32 k3 s1)
Angular Feature Extraction	reshape	Space to Viewmap
	I0	conv2d(n128 k15 s15)
	I1	conv2d(n128 k7 s1)
	I2	subpixel(r=2)->conv2d(n64 k3 s1)
Feature Fusion	I3	subpixel(r=2)->conv2d(n32 k3 s1)
	concat	concat (I2,I3)
U-Net (Encoder)	I0	conv2d(n32 k3 s1)->InstanceNorm->LeakyRelu
	En_0	conv2d(n64 k3 s1)->InstanceNorm->LeakyRelu
	I0	conv2d(n128 k1 s1)->InstanceNorm->LeakyRelu
	I1	conv2d(n32 k3 s1)->InstanceNorm->LeakyRelu
	I2	conv2d(n32 k3 s1)->InstanceNorm->LeakyRelu
	I3	conv2d(n64 k3 s1)->InstanceNorm->LeakyRelu
	I4	concat(I1,I2,I3)->InstanceNorm
	merge(I0,I4)	merge(I0,I4)->LeakyRelu->InstanceNorm
	En_1	Maxpooling (k3 s2)
	I0	conv2d(n256 k1 s1)->InstanceNorm->LeakyRelu
	I1	conv2d(n64 k3 s1)->InstanceNorm->LeakyRelu
	I2	conv2d(n64 k3 s1)->InstanceNorm->LeakyRelu
	I3	conv2d(n128 k3 s1)->InstanceNorm->LeakyRelu
	I4	concat(I1,I2,I3)->InstanceNorm
	En_2	Maxpooling (k3 s2)
	I0	conv2d(n512 k1 s1)->InstanceNorm->LeakyRelu
	I1	conv2d(n128 k3 s1)->InstanceNorm->LeakyRelu
	I2	conv2d(n128 k3 s1)->InstanceNorm->LeakyRelu
	I3	conv2d(n256 k3 s1)->InstanceNorm->LeakyRelu
	I4	concat(I1,I2,I3)->InstanceNorm
	En_3	Maxpooling (k3 s2)
	I0	conv2d(n512 k1 s1)->InstanceNorm->LeakyRelu
	I1	conv2d(n128 k3 s1)->InstanceNorm->LeakyRelu
	I2	conv2d(n128 k3 s1)->InstanceNorm->LeakyRelu
	I3	conv2d(n256 k3 s1)->InstanceNorm->LeakyRelu
	I4	concat(I1,I2,I3)->InstanceNorm
	En_4	Maxpooling (k3 s2)
	I0	conv2d(n512 k1 s1)->InstanceNorm->LeakyRelu
	I1	conv2d(n128 k3 s1)->InstanceNorm->LeakyRelu
	I2	conv2d(n128 k3 s1)->InstanceNorm->LeakyRelu
	I3	conv2d(n256 k3 s1)->InstanceNorm->LeakyRelu
	I4	concat(I1,I2,I3)->InstanceNorm->Maxpooling (k3 s2)
U-Net (Middle)	Middle	Upsamling concat(En_4,Middle)
U-Net (Decoder)	DE_1	conv2d(n512 k3 s1)-> LeakyRelu->InstanceNorm->Upsamling
	DE_2	concat(En_3,DE_1)->conv2d(n512 k3 s1)-> LeakyRelu->InstanceNorm->Upsamling
	DE_3	concat(En_2,DE_2)->conv2d(n256 k3 s1)-> LeakyRelu->InstanceNorm->Upsamling
	DE_4	concat(En_1,DE_3)->conv2d(n128 k3 s1)-> LeakyRelu->InstanceNorm->Upsamling
	DE_5	concat(En_0,DE_4)->conv2d(n161 k3 s1)-> LeakyRelu->InstanceNorm->Upsamling
Refine & Output	conv_final	conv2d(slices k3 s1)